

Smart Wallet with Budget Guardian

Business Requirements Document

Project Group Number: 14
Tutorial Group: 1

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1 Product Scope & Overview

System Definition

The **Smart Wallet with Budget Guardian** is a database-driven personal finance management application. Unlike standard digital wallets, this system is engineered to facilitate digital payments while strictly enforcing financial discipline through active monitoring of global assets and liabilities.

The core functionality allows users to maintain a digital wallet, perform real-time transactions, simulate cryptocurrency investments, and ensure rigorous tracking of monthly spending against self-defined budgets using automated database triggers.

2 Problem Statement & Business Purpose

The Problem

Traditional expense trackers are "passive"—they only log expenses after the transaction occurs. This retrospective approach fails to prevent "lifestyle creep" or impulsive spending in real-time.

Business Purpose

The purpose of this project is to shift from passive recording to **active enforcement**. We are developing a "Budget Guardian" engine that utilizes database-level constraints to monitor spending *transaction-by-transaction*.

The goal is to provide a unified platform that not only tracks liquid cash and digital assets but acts as a financial safeguard, alerting users immediately when they breach their financial limits.

3 Stakeholder Analysis & Use Cases

3.1 Primary Stakeholders

- **The End User (Account Holder):** Students or young professionals managing personal finances, daily expenses, and crypto investments.
- **System Administrator:** Responsible for maintaining global system data (e.g., Exchange Rates, Expense Categories).
- **External Financial Entity:** The simulated bank providing funds to the digital wallet.

3.2 Key Use Cases

1. Wallet Loading:

- *Actor:* User
- *Action:* Transfer funds from linked Bank Account to Wallet.

2. Budget Definition:

- *Actor:* User
- *Action:* Set a monthly limit (e.g., \$500) for a specific Category (e.g., "Dining").

3. Transaction Execution with Guardian Check:

- *Actor:* User

- *Action:* Initiate payment for a service.
- *System Logic:* The system deducts balance AND checks if the new total exceeds the budget. If yes, an Alert is generated.

4. Asset Diversification:

- *Actor:* User
- *Action:* Swap liquid wallet balance for Crypto Assets to track Net Worth.

4 Functional Requirements

4.1 User Registration & Wallet Management

- **Registration:** A user must register on the platform to create a unique Wallet. Required details include name, email, and password.
- **Bank Account Linkage:** Users must link a simulated Bank Account to their wallet to serve as the source for loading funds.
- **Single Wallet Policy:** To ensure financial data integrity, each user is linked to exactly one wallet.
- **Multi-Currency Display:** Financial records are stored in a base currency (e.g., USD), but users may toggle their preferred view (e.g., EUR, JPY).

4.2 Transactions & Categorization

- **Execution & Recording:** Users perform transactions to pay for goods or services. The system records the amount, type (Credit, Debit, or Trade), and timestamp.
- **Categorization:** Every transaction is tagged with a Category (e.g., Food, Utilities) and a Priority Level (Essential, Flexible, Luxury).

4.3 Crypto Asset Tracker

- **Asset Purchase:** Users can use their wallet balance to "buy" virtual cryptocurrency assets.
- **Net Worth Calculation:** The system calculates total Net Worth by combining the liquid wallet balance with the current market value of crypto holdings.

4.4 Budget Guardian & Alerts

- **Guardian Mechanism:** An automated mechanism monitors spending. If spending exceeds the defined limit, the system immediately generates an Alert.
- **Notification History:** Alerts notify the user of the breach and are stored for historical reference.

5 Key Entities

1. **User:** Individual registered on the platform who owns a single wallet.
2. **Bank Account:** Simulated external account required for funding the wallet.
3. **Wallet:** Digital repository holding the user's liquid cash balance.
4. **Transaction:** Financial event recording movement of funds.
5. **Category:** Classification tag for expenses assigned a specific priority level.
6. **Budget:** User-defined financial limit linked to a category.
7. **Alert:** Notification generated when a user breaches their budget.
8. **Crypto Asset:** Digital asset holding purchased using wallet funds.

6 System Consistency & Integrity

- **Automated Updates:** When a debit transaction occurs, the wallet balance decreases and the budget's "spent" amount increases simultaneously.
- **Validation:** If the balance is insufficient for a payment or asset purchase, the transaction is flagged or blocked.
- **Referential Integrity:** The database design ensures all records (users, wallets, transactions, budgets) are linked correctly to prevent data discrepancies.